

1a)

I randomly selected 3 rows from my dataframe and printed out the rows to get the review_body and star_rating values.

Then, I obtained rating statistics using pandas' value_counts() function on the star_rating column.

Sample Review	Rating
Great product.	5
Gold plated fusers are the best! It will never fail on you	5
Gorgeous colors and easy to use	4

Rating	Count
5	1580941
4	417975
3	193452
2	138215
1	306576

1b)

I first created a new "sentiment" column with default value of -1. Then, I used df.loc to set sentiment of all reviews with star_rating ≥ 4 to 1 and star_rating ≤ 2 to 0.

Sentiment	Count
Positive (1)	1998916
Negative (0)	444791
Neutral (-1)	193452

2)

I performed data cleaning using a combination of pandas builtin functions, BeautifulSoup and regex. I defined as many contractions as I could think of, and additionally added mappings for contractions without apostrophes as I expect many people to write contractions the lazy way in their reviews.

	Average Character Length
Before cleaning	319
After cleaning	305

3)

I removed stopwords using nltk's set of English stopwords and excluded words like will, would, etc. as well as no and not, as they are important to preserving sentiment. I then performed lemmatization using nltk's WordNetLemmatizer

Review comparison before/after data cleaning + preprocessing	
Before	After
I wish I had read the reviews before I purchased them. It didn't ruin the printer but they didn't work and I needed to get some photos printed. I am extremely disappointed.	wish have read review purchase do not ruin printer do not work need get photos print be extremely disappoint
The day of the week is automatically populated when you enter the date. But it is set for 2008, and there is no way to change it to the correct day of the week. I set it on Saturday, and there was no way to get it to say Saturday. It will only have the correct day of the week once every seven years. I don't see any advantage to the magnetic key holders over simple hooks. Other than you are forced to put your keys on the magnetic key ring that comes with the product. Because we all need one more thing in our pockets. I wouldn't buy this again.	day week be automatically populate enter date be set be no way change correct day week set saturday no way get say saturday will have correct day week every seven years do not see advantage magnetic key holders simple hook be force put key magnetic key ring come product need one thing pocket would not buy

Lasted 2 weeks and no longer works	last weeks no longer work
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	Average Character Length
Before preprocessing	305
After preprocessing	201

5) Perceptron

Perceptron Training Accuracy: 0.9948

Perceptron Training Precision: 0.9908

Perceptron Training Recall: 0.9990

Perceptron Training F1-score: 0.9949

Perceptron Testing Accuracy: 0.8930

Perceptron Testing Precision: 0.8934

Perceptron Testing Recall: 0.8923

Perceptron Testing F1-score: 0.8929

All 4 test metrics have approximately the same value for the perceptron model.

6) SVM

SVM Training Accuracy: 0.9952

SVM Training Precision: 0.9908

SVM Training Recall: 0.9997

SVM Training F1-score: 0.9952

SVM Testing Accuracy: 0.8837

SVM Testing Precision: 0.8639

SVM Testing Recall: 0.9109

SVM Testing F1-score: 0.8867

SVM resulted in higher recall than precision, meaning it produced fewer false negatives but more false positives.

7) Logistic Regression

Logistic Regression Training Accuracy: 0.9887
Logistic Regression Training Precision: 0.9804
Logistic Regression Training Recall: 0.9975
Logistic Regression Training F1-score: 0.9888
Logistic Regression Testing Accuracy: 0.8985
Logistic Regression Testing Precision: 0.8767
Logistic Regression Testing Recall: 0.9276
Logistic Regression Testing F1-score: 0.9014

Logistic regression resulted in higher recall than precision, meaning it produced fewer false negatives but more false positives.

8) Multinomial Naive Bayes

Naive Bayes Training Accuracy: 0.9457
Naive Bayes Training Precision: 0.9794
Naive Bayes Training Recall: 0.9106
Naive Bayes Training F1-score: 0.9438
Naive Bayes Testing Accuracy: 0.8688
Naive Bayes Testing Precision: 0.8870
Naive Bayes Testing Recall: 0.8452
Naive Bayes Testing F1-score: 0.8656

Naive Bayes resulted in lower recall than precision, meaning it produced more false negatives but fewer false positives.